DATA SCIENCE - 22DS102006

1. Data Wrangling
   a. Perform hierarchical indexing by creating a series with a list of lists (or arrays) as the index, select subsets of data at outer and inner levels using partial indexing.
   b. Rearrange the tabular data with hierarchical indexing using unstack and stack method.
   c. Create two different DataFrames and merge them using index as merge key and combine data with overlap using combine_first method.

4. Data Wrangling

a. Hierarchical Indexing

```
import pandas as pd
import numpy as np

data = pd.Series(np.random.randn(6),
index=[['a', 'a', 'b', 'b', 'c', 'c'],
       [1, 2, 1, 2, 1, 2]])

print(data)
print(data['b'])
print(data[:, 1])
```

b. Stack and Unstack

```
df = data.unstack()
print(df)
print(df.stack())
```

c. Merge and Combine

```
df1 = pd.DataFrame({'a': [1, np.nan, 3], 'b': [4, 5, 6]})
df2 = pd.DataFrame({'a': [7, 8, 9], 'b': [np.nan, 11, 12]})

merged = df1.combine_first(df2)
print(merged)
```